FOLLOWER

ROVER

AUTONOMOUS PROTOTYPE

Advanced Human-Robot Interaction
& Tracking System — Slide Deck
Project Version 1.0

AUTONOMOUS MAN-FOLLOWING SYSTEM

Designing a robotic platform that seamlessly identifies and follows a human target in dynamic environments. A deep dive into sensing, control, and safety frameworks.

PLATFORM	ROBOTIC ROVER	10 SLIDES

Exploring sensor fusion, person identification, PID motor control, and collision avoidance strategies for real-world assistive robotics.

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ROVER GUIDE

02



INTENTIONAL | SYSTEMATIC | HUMAN-LED

EVERY MOVE IS A CALCULATED RESPONSE.

The follower rover operates on a continuous feedback loop that synchronizes mechanical motion with human intent. It is not just a machine moving; it is a system perceiving, predicting, and performing.

By maintaining a strict distance-to-speed ratio, the prototype ensures that the human target is always within safe reach but never obstructed, creating a seamless collaborative experience.

THE CORE PHILOSOPHY

THE FOLLOW LIFECYCLE

The rover maintains proximity through a high-frequency loop of sensing and motion correction.

01

DETECTION

Scanning environment for human infrared or visual signature.

DISCOVER

02

ANALYSIS

Calculating distance and deviation from center heading.

DEFINE

03

MOTION

Executing smooth acceleration and steering maneuvers.

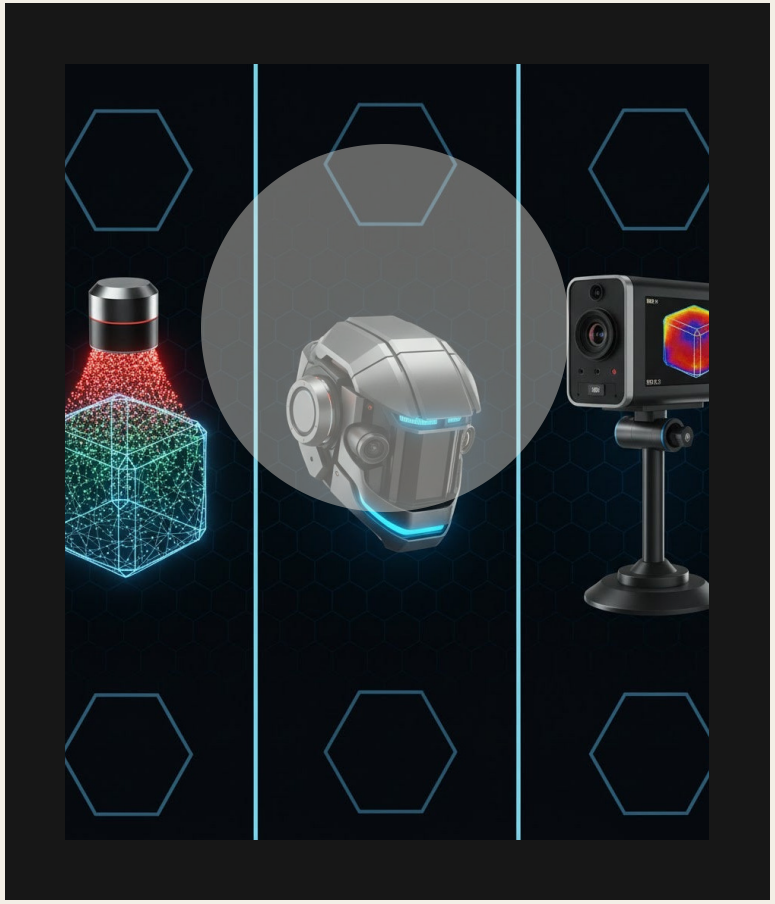
DELIVER

04

FEEDBACK

Verifying target position and adjusting for noise.

ITERATE



VISION & PERCEPTION

IDENTIFICATION

The system uses a combination of ultrasonic ranging and infrared signatures to lock onto the human target, filtering out background noise and static obstacles in real-time.

PERSISTENCE

Advanced tracking algorithms maintain target lock even during brief occlusions or sharp turns, utilizing predictive motion estimation to keep the rover on path.

DETECTION & TRACKING

HARDWARE ARCHITECTURE

THE INTEGRATED SYSTEM

The hardware stack is designed for modularity and high-speed processing. At the core, an embedded microcontroller fuses data from multiple sensor inputs to create a 360-degree awareness map.

Reliable motion is achieved through high-torque DC motors equipped with optical encoders, ensuring precise speed control and feedback even on uneven terrain.

01

SENSORS

LiDAR & Ultrasonic depth sensing.

02

CONTROLLER

MCU-based PID logic.

03

DRIVE

Dual-motor differential steering.

04

POWER

Lithium management system.

FOUR PILLARS OF CONTROL

The control algorithm ensures that the rover behaves predictably. By balancing these four pillars, we achieve a following experience that feels natural and safe for the user.

01 **PROXIMITY**
Maintaining a consistent 1.0m to 1.5m buffer zone from the target.

02 **ORIENTATION**
Active steering adjustment to keep the human target centered.

03 **VELOCITY**
Dynamic acceleration matching the user's natural walking speed.

04 **SAFETY**
Emergency override for immediate halt when obstacles are detected.

The fusion of distance sensing and motor response creates a stable, intuitive following behavior.

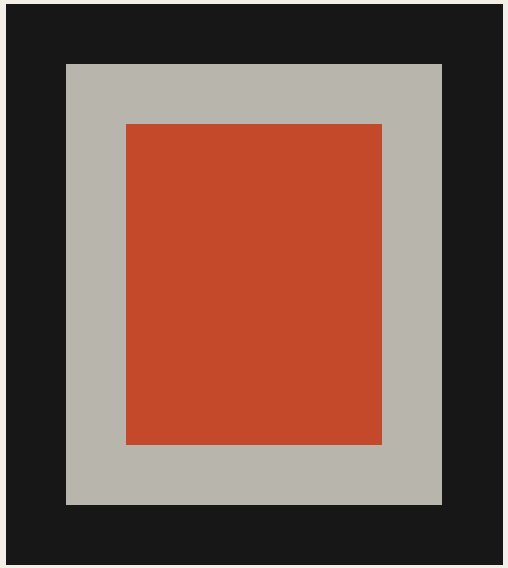
MEASURE PERFORMANCE

<50ms
LATENCY

Processing delay from sensor input to motor action, ensuring real-time responsiveness.

±5cm
ACCURACY

Precision in maintaining the set buffer distance from the human target during motion.



98%
UPTIME

Reliability of target acquisition in varying light and acoustic conditions.

100%
SAFETY

Successful emergency stops when obstacles appear or when the target is lost.

WHERE IT HELPS

From industrial logistics to personal care, the man-following rover addresses the "last meter" of transport in diverse environments.

LOGISTICS

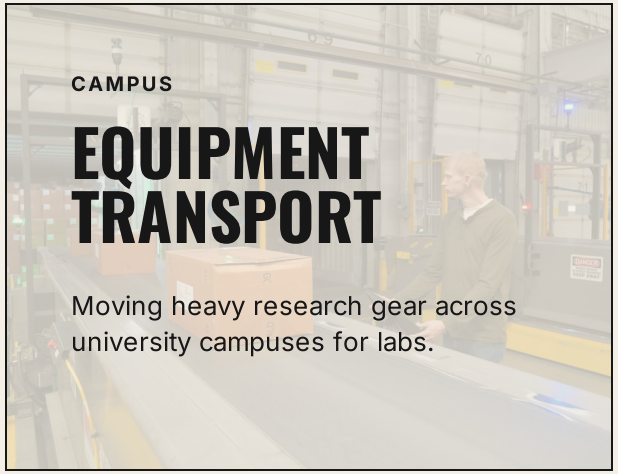
WAREHOUSE FULFILLMENT

Assisting workers by carrying heavy loads in large fulfillment centers.

MEDICAL

HOSPITAL ASSISTANT

Transporting medical supplies between wards following nursing staff.



CAMPUS

EQUIPMENT TRANSPORT

Moving heavy research gear across university campuses for labs.

DOMESTIC

PERSONAL AIDE

Following the elderly at home with essential items and emergency kits.

PRIMARY USE CASES

INDUSTRIAL HEALTHCARE CAMPUS ELDERLY CARE

NEXT STEPS

1. **BENCH TESTING**

   Validating sensor fusion accuracy and PID loop tuning in lab settings.

2. **FIELD TRIALS**

   Supervised operation in indoor environments with moving human targets.

3. **DEPLOYMENT**

   Full-scale integration into warehouse logistics and fulfillment workflows.

LET'S BUILD THE ROVER

The future of robotics is collaborative. Join us in refining the next generation of autonomous following systems.

STRATEGY IDENTITY EXECUTION

